AI 101 for Mechanical Engineers · Module 1

A map of related ideas

Lesson 1.1

What you will learn

> Explain how AI, machine learning, deep learning, generative AI and data science relate, and what each does in an engineering system.

Lesson overview

Module 1 · Lesson 1

A map of related ideas

AI 101 for Mechanical Engineers

1 How the ideas fit

Artificial intelligence (AI)
Includes rules, search and learning

Machine learning (ML)
Learns from data

Deep learning
Uses neural networks with multiple layers

AI does not always learn from data.

2 What does the model produce?

Classification
Predicts a category

Surface image → Classifier → Cracked or intact

Generation
Produces content

Maintenance information → Text generator → Draft explanation

Deep learning describes the approach. Generation describes the output.

3 Different jobs need different checks

Component	What to check
Fixed temperature trip	Setpoint and response time
Bearing-fault classifier	Performance on independent examples
Maintenance draft	Records and calculations

A well-written draft can still contain errors.

4 Data science includes the whole workflow

Measure → **Clean** → **Analyse** → **Model** → **Communicate**

Check sensor calibration and missing readings before relying on model outputs.

AI 101 for Mechanical Engineers · Lesson 1.1

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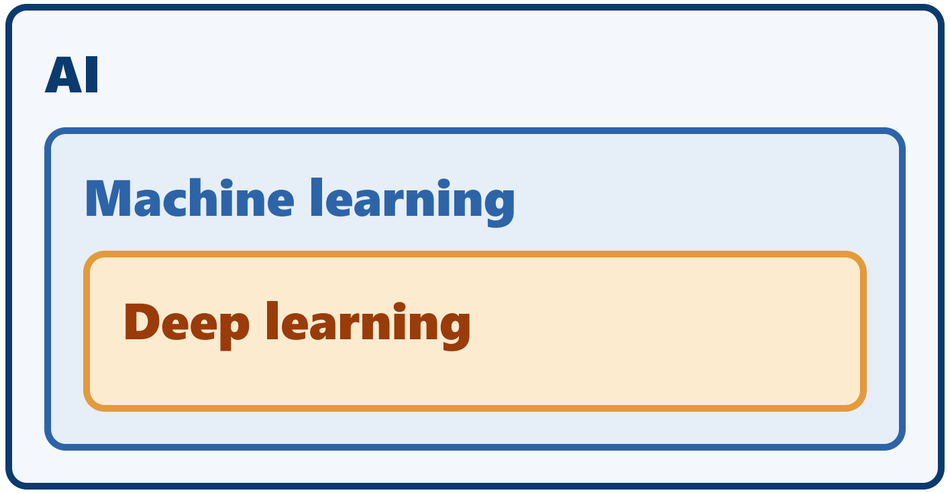
Engineers stay responsible for the decision

- Measurements: temperature, pressure, vibration, flow, current
- AI and ML can find patterns or do a defined task
- Engineers still check results against the physical system
- Ask what evidence you need before acting

AI is a broad field

- Systems doing tasks linked to intelligent behaviour [1]
- Some follow fixed rules written in advance
- Others search, use stored knowledge or learn from examples
- “AI” alone does not tell you whether it learns

ML is inside AI; deep learning is inside ML



- ML learns patterns or parameters from examples [2]
- Deep learning: ML with multi-layer neural networks [3], [4]
- Not all AI uses neural networks

Approach and output are different questions

Classification

- Predicts a category
- e.g. casting image: cracked or intact

Generation

- Produces new content [4, Ch. 20]
- e.g. drafts a maintenance explanation

> A deep-learning model can be either. Generative AI is not a fourth nested level.

Data science covers the whole workflow



- Modelling is only one stage [5]
- Check sensor calibration and missing readings first

One plant, three components, three checks

Component	Produces	Check
Over-temperature trip	Trip command (fixed rule)	Setpoint and response time
Bearing-fault classifier	Fault estimate (learned)	Independent test examples
Generative assistant	Maintenance draft	Records and calculations

CHECK YOUR UNDERSTANDING

Can a deep-learning system be non-generative?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Can a deep-learning system be non-generative?

> Yes. A multi-layer network can label a surface image as cracked or intact without generating new content.

CHECK YOUR UNDERSTANDING

Why is generative AI not a compulsory level inside deep learning?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why is generative AI not a compulsory level inside deep learning?

> Generation describes the output. Deep learning describes the model approach. They answer different questions.

Key takeaways

- AI includes learning and non-learning approaches
- ML is within AI; deep learning is ML with multi-layer networks
- Classification predicts a category; generation produces content
- Data science spans measurement to communication
- Match each component’s output to the right check

References

[1] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. (US). Hoboken, NJ, USA: Pearson, 2021. ISBN 978-0-13-461099-3. (The authors date this edition 2020.)

[2] K. P. Murphy, Probabilistic Machine Learning: An Introduction. Cambridge, MA, USA: MIT Press, 2022.

[3] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” Nature, vol. 521, no. 7553, pp. 436–444, 2015. doi: 10.1038/nature14539

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

[5] F. Provost and T. Fawcett, “Data science and its relationship to big data and data-driven decision making,” Big Data, vol. 1, no. 1, pp. 51–59, 2013. doi: 10.1089/big.2013.1508

Numbers match the course reference list. The written lesson is the authoritative text.